Jafer Sathik B

Software Developer | Full-Stack & AI/ML
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Professional Summary

B.Tech Information Technology student and software developer who designs, builds and deploys production web applications and AI/ML systems end to end. Experienced in full-stack development, database design, role-based security, automated testing and Linux deployment, and in leading a client project from requirements to release. Works across Python, JavaScript, PHP and SQL, and picks up new tools quickly.

Education

PSG College of Technology, Coimbatore 2024 – 2027
B.Tech in Information Technology

Experience

Custom Inventory Management System Dec 2025 – Apr 2026
Project Lead / Full-Stack Developer – Client Project *IT Core Solutions, Coimbatore*
Live Demo | Certificate
Tech Stack: PHP, Laravel, MySQL, Blade, JavaScript, Tailwind CSS, PHPUnit, Telegram Bot API, Linux, cPanel

- Architected and deployed a manufacturing inventory platform with BOM calculations (including composite materials), order management, stock control, stock history and PDF reporting on Linux/cPanel servers.
- Prevented negative stock under simultaneous orders by running the stock check and deduction in one **database transaction with row locking**.
- Added role-based access, Telegram alerts, daily PDF reports and a live stock-check preview, backed by a **36-test automated suite** covering orders and permissions.

Yutira 2026 Event Management Platform Mar 2026
Freelance Full-Stack Developer *Dept. of Civil Engineering, PSG College of Technology*
Live Demo | Certificate
Tech Stack: Next.js, React, JavaScript, MongoDB, Mongoose, JWT, Nodemailer, Tailwind CSS

- Built a registration platform and admin portal with email verification, unique participant IDs, payment tracking, attendance marking, paper review and CSV export.
- Ran live operations at a two-day symposium: **248 registrations and 28 paper submissions** with zero downtime.

Projects

Privacy-Preserving RAG
Tech Stack: Python, FastAPI, FAISS, Sentence Transformers, Ollama (Llama 3.1), OCR, Merkle Trees

- Built a document Q&A system that retrieves over **masked query embeddings** with clustered FAISS indexes, so raw embeddings are never stored.
- Added **Merkle-tree verification** of retrieved passages, fail-closed answer validation and verified citations in a streaming chat interface.
- Sped up queries with incremental indexing and cluster routing; fixed path-traversal, XSS and unsafe `eval` issues.

Predictive Vehicle Maintenance
Tech Stack: Python, Pandas, NumPy, scikit-learn, Random Forest, XGBoost, LightGBM, SHAP, Streamlit

- Built an ML tool that predicts which vehicles need maintenance from usage, condition and service-history features, replacing fixed service schedules.
- Compared Random Forest, XGBoost and LightGBM, added **SHAP explainability**, and served single and batch CSV predictions through Streamlit.

Government Plot Compliance Web Application
Niral Thiruvizha 3.0
Tech Stack: PHP, Laravel, MySQL, Blade, Tailwind CSS, RBAC, DomPDF

- Built a Laravel app covering the full plot-lease lifecycle: allotment, milestone compliance, incentive claims, closure certificates and legal escalation.
- Added a daily scheduler for deadline and overdue alerts, **role-based access** limiting each allottee to their own cases, audit logging and PDF reports.

Technical Skills

Languages: Python, JavaScript, TypeScript, PHP, SQL, HTML, CSS
Web Development: React, Next.js, Laravel, FastAPI, Blade, Tailwind CSS, REST APIs, Vite
Databases & Data: MySQL, MongoDB, Pandas, NumPy, scikit-learn
AI / ML: RAG, LLMs (Ollama), FAISS, Sentence Transformers, XGBoost, LightGBM, Random Forest, SHAP
DevOps, Testing & Security: Git, Docker, Linux, cPanel, PHPUnit, JWT, bcrypt, RBAC, Merkle Trees

Achievements & Leadership

- **Hackathons:** Selected at college level in **Smart India Hackathon 2024**; selected among the **Top 1,000 ideas** in Round 1 of Niral Thiruvizha 3.0.
- **Academic Excellence:** School First (**98%**), Diploma Department First (**95%**).
- **Leadership:** Project Lead for client inventory software, Panel Head for 7th-semester project reviews, IEEE Association Media Team Member.